A

PROJECT REPORT ON

"MAHADISCOM WEBSITE AUTOMATION USING UIPATH"

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

BACHELOR OF ENGINEERING (Computer Engineering)

BY

MR. KHOPADE MANISH JEEVAN. EXAM NO: - B400990091
MR. GHANGALE SANGRAM BHARAT. EXAM NO: - B400990078
MS. PUNDE VAISHNAVI EKNATH. EXAM NO: - B400990101
MS. TIKANDE DIVYA DADAPATIL. EXAM NO: - B400990099

UNDER THE GUIDANCE OF

Prof. K. B. Khatal

DEPARTMENT OF COMPUTER ENGINEERING
SAHYADRI VALLEY COLLEGE OF ENGINEERING & TECHNOLOGY
Rajuri, Kalyan - Nagar Highway, Taluka - Junnar, District - Pune

2024-2025

SAHYADRI VALLEY COLLEGE OF ENGINEERING & TECHNOLOGY, RAJURI

DEPARTMENT OF COMPUTER ENGINEERING

Rajuri, Kalyan - Nagar Highway, Taluka - Junnar, District - Pune

CERTIFICATE

This is to Certify that project stage 1 report entitled

"MAHADISCOM WEBSITE AUTOMATION USING UIPATH"

SUBMITTED BY

BE-002

Is a bonafide work carried out by her under the supervision Prof. K. B. Khatal and it is submitted towards the partial fulfillment of the requirement of University of Pune. For the award of the degree of bachelor of Engineering (Computer Engineering)

Internal Examiner	External Examiner
(Prof. K. B. Khatal)	(Prof.)
Head Of Department	Principal
(Prof. K. B. Khatal)	(Prof. P. Balaramudu)

Savitribai Phule Pune University

CERTIFICATE

This is to Certify that

Mr. Khopade Manish Jeevan
Mr. Ghangale Sangram Bharat
Ms. Vaishnavi Eknath Punde
Ms. Tikande Divya Dadapatil

Student of B.E. Computer (Second Year) was examined in

Project Project Stage 2 entitled

"MAHADISCOM WEBSITE AUTOMATION USING UIPATH"

on .../... /2025

At

Department of Computer Engineering,

SAHYADRI VALLEY COLLEGE OF ENGINEERING,

Rajuri - 412411

.....	.....
Internal Examiner	External Examiner
(Prof. K. B. Khatal)	(Prof.)

Certificate By Guide

This is to certify that **(BE-002) Mr. Khopade Manish Jeevan Mr. Ghangale Sangram Bharat, Ms. Tikande Divya Dadapatil, Ms. Vaishnavi Eknath Punde** has completed the Project Stage 2 work under my guidance and supervision and that, I have verified the work for its originality in documentation, problem statement, implementation and results presented in the Project Stage 2. Any reproduction of other necessary work is with the prior permission and has given due ownership and included in the references.

Place: Rajuri

Date: (Prof. K. B. Khatal)

ACKNOWLEDGEMENT

"Feeling gratitude and not expressing it, is like wrapping a present and not giving it"

The satisfaction that accompanies the successful completion of any task would be incomplete without mentioning the people who made it possible. So, I am grateful to number of individuals whose professional guidance along with encouragement have made it very pleasant endeavor to undertake this project.

I take this opportunity to express my profound gratitude and deep regards to my guide and truly teacher of teachers **Prof. K. B. Khatal** for their exemplary guidance, monitoring and constant encouragement throughout the course of this project. I also express a deep sense of gratitude to **Prof. K. B. Khatal**, Head of Computer Engineering Department for their valuable guidance and encouragement.

I am obliged to our Principal **Dr S. B. Zope** and vice Principal **Prof P. Balaramudu** for their inspiration and co-operation.

Last but not least I have to express our feelings towards all staff members of Sahyadri Valley College Of Engineering and special thanks to my family, colleague and friends for their moral support and help.

Mr. Khopade Manish Jeevan.
Mr. Ghangale Sangram Bharat.
Mr. Vaishnavi Eknath Punde.
Mr. Tikande Divya Dadapatil.
SVCET,Rajuri.

LIST OF PUBLICATIONS

1. Published paper entitled *'Data Capturing in Web Applications Using RPA UiPath"*, This paper outlines a systematic web-based approach to facilitate data collection using UiPath, providing a detailed example of a web-based method developed for a study in Starr County, Texas.
   Download Link : $https://ijrpr.com/uploads/V4ISSUE6/IJRPR14561.pdf$

2. Published paper entitled *"MAHADISCOM WEBSITE AUTOMATION USING UIPATH"*
   Download Link : $https://ijrpr.com/uploads/V4ISSUE6/IJRPR14561.pdf$

3. Published paper entitled *"A Systematic Review of Robotic Process Automation in Business Operations: Contemporary Trends and Insights"*, in This paper provides a systematic review of RPA in business operations, highlighting contemporary trends and insights.
   Download Link : $https://www.researchgate.net/publication/3738458$

4. Paper presented on *"Enabling Digital Transformation in Government with Automation"*, This article discusses how RPA is enabling digital transformation in government agencies and how automation resolves significant problems facing government institutions.

ABSTRACT

In the digital era, government service portals like MAHADISCOM (Maharashtra State Electricity Distribution Company Limited) play a crucial role in delivering essential electricity-related services to millions of citizens. However, the current processes involved in accessing these services—such as bill payments, new connection applications, complaint registrations, and service tracking—are often manual, repetitive, and time-consuming. These manual tasks not only consume a significant amount of human effort but also lead to delays, data entry errors, and poor customer experience. To address these challenges, this project proposes the implementation of Robotic Process Automation (RPA) using UiPath, a powerful automation tool designed to mimic human interactions with digital systems. The aim of this project is to automate key operations on the MAHADISCOM website, thereby reducing manual workload, increasing process accuracy, and speeding up service delivery. The automation bots are designed to interact with the website interface, extract relevant data, fill out forms, download bills, submit requests, and generate automated logs or reports. The project also focuses on ensuring data security and compliance while interacting with public-facing systems. By integrating UiPath with MAHADISCOM's workflow, the project demonstrates how automation can transform public utility services by improving efficiency, reducing human dependency, and paving the way for a smarter, faster, and more responsive digital governance model. This initiative serves as a practical case study on how RPA can be effectively implemented in public sector utilities for digital transformation and service excellence.

Keywords - Electricity bill payments, New connection requests, Consumer complaint registration, and Data extraction

INDEX

Acknowledgement **i**

List of Publications **ii**

Abstract **iii**

Index **iv**

Synopsis **vi**

1 INTRODUCTION **2**

2 OBJECTIVES **4**

2.1 Specific Objectives: 4

3 LITERATURE SURVEY **6**

3.1 Role of RPA in Automating Web-Based Systems 6

3.2 UiPath as a Preferred RPA Tool 6

3.3 RPA in Public Sector Applications 7

3.4 Automation of Utility Services 7

3.5 Challenges and Best Practices in RPA Deployment 7

3.6 Web Scraping and Data Extraction Techniques 7

4 Problem Statement **9**

5 Necessity of the Project **11**

6 Existing System **13**

7 Proposed System **15**

8 System Architecture **17**
8.1 Workflow of the System Architecture: 18

9 Software and Hardware Requirements **19**
9.1 Software Requirements 19
9.2 Hardware Requirements 20
9.3 Other Requirements 20
9.4 Environment Setup Notes 20

10 Mathematical Model **21**

11 Project Planning **24**
11.1 Project Stage 2 Title 24
11.2 Project Duration 24
11.3 Project Milestones 24
11.4 Tools Technologies 25
11.5 Success Criteria 25

12 Advantages **26**

13 Real-Time Applications **28**

14 Conclusion **30**

15 Future Scope **31**

References **33**

SYNOPSIS

Project Stage 2 Title: "MAHADISCOM Website Automation Using UiPath".

Internal Guide: Prof. K. B. Khatal

Existing System

MAHADISCOM (Maharashtra State Electricity Distribution Company Limited) is one of the largest electricity distribution utilities in India, serving millions of consumers across Maharashtra. The official MAHADISCOM website offers various consumer services such as bill payments, new connection applications, complaint registration, and status tracking. However, these services often require manual effort for repetitive web interactions, which can lead to delays, human errors, and inefficiencies. In today's digital age, there is a growing need for automation in such government portals to improve operational efficiency and customer satisfaction.

Motivation

- In today's fast-paced digital world, both private and public sector organizations are seeking ways to improve service delivery, reduce operational costs, and enhance user satisfaction. MAHADISCOM, being one of the largest electricity distribution companies in India, caters to millions of consumers daily. However, many of its routine operations—such as bill viewing, complaint registration, and new connection applications—are still handled manually by users, which is time-consuming, error-prone, and inefficient. These repetitive tasks lead to delays in service delivery and increased workload for employees.

- Observing this gap in efficiency and the growing demand for seamless digital interactions inspired the motivation to explore Robotic Process Automation (RPA) as a solution. RPA, especially with tools like UiPath, has the capability to automate repetitive web-based tasks without changing the existing IT infras-

tructure. Automating the MAHADISCOM website processes using UiPath not only improves the speed and accuracy of operations but also contributes to the larger goal of digital transformation in public utilities.

- This project is driven by the desire to implement a real-world automation solution that can bring measurable benefits to both users and service providers. It is also motivated by the potential of RPA to transform how government services are delivered in India, making them faster, smarter, and more reliable. By taking this initiative, we aim to contribute toward building a more efficient and citizen-friendly digital ecosystem.

Problem Statement:

MAHADISCOM (Maharashtra State Electricity Distribution Company Limited) provides essential electricity services to millions of consumers across the state. The organization's website serves as a central platform for various consumer services such as bill payments, complaint registrations, service requests, and status tracking. However, the current process of accessing and managing these services through the website is largely manual, repetitive, and time-consuming. Users and staff must repeatedly input data, navigate multiple pages, and perform routine tasks which increases the risk of human error, delays, and inconsistent service delivery.

Additionally, for internal employees handling a large volume of consumer data and service requests daily, the workload becomes overwhelming and inefficient. These manual tasks not only slow down the response time but also result in reduced productivity and poor customer experience. There is a clear need for a solution that can automate these repetitive processes, minimize human intervention, and improve operational efficiency.

This project addresses the problem by implementing Robotic Process Automation (RPA) using UiPath to automate interactions with the MAHADISCOM website. The automation will enable tasks such as automatic login, data extraction, bill downloading, complaint registration, and service request submissions to be handled by software robots, thereby reducing manual effort, increasing accuracy, and improving overall system performance. The goal is to develop a scalable, cost-effective, and efficient automation system that enhances the digital experience for both MAHADISCOM employees and consumers.

Solving Approach:

To solve the challenges associated with the manual processing of tasks on the MAHADISCOM website, this project adopts a structured Robotic Process Automation (RPA) approach using UiPath, a leading automation platform. The proposed solution is designed to replace repetitive, time-consuming, and error-prone manual activities with intelligent automation workflows. The solving approach is divided into multiple stages to ensure clarity, scalability, and efficiency.

Keywords:

1. **MAHADISCOM (Maharashtra State Electricity Distribution Company Limited):**

   A government-operated utility that provides electricity to millions of consumers in Maharashtra. The automation is specifically focused on its official service website.

2. **Robotic Process Automation (RPA):**

   A technology that uses software robots or "bots" to mimic human actions on digital interfaces. In this project, RPA is used to automate tasks like bill payment and complaint registration.

3. **UiPath:**

   A widely used RPA tool that provides a visual environment (UiPath Studio) to create automation workflows for repetitive tasks. It enables drag-and-drop development, web automation, data extraction, and more.

4. **Website Automation:**

   The process of using software to interact with a website's interface—such as filling out forms, clicking buttons, or reading data—without human involvement. This is the core task being performed on the MAHADISCOM website.

5. **Electricity Bill Payment:**

   One of the most common user activities on the MAHADISCOM portal. The automation bot can retrieve billing information and send payment reminders or automate billing workflows.

Chapter 1

INTRODUCTION

In today's digital-first world, automation has emerged as a transformative solution for improving the efficiency, reliability, and scalability of business operations across various sectors—including the public utility domain. Government services, which often handle massive volumes of public data and citizen requests, are gradually adopting automation to overcome challenges like limited human resources, process delays, and operational inefficiencies. One such example is MAHADISCOM (Maharashtra State Electricity Distribution Company Limited), a major public sector undertaking responsible for electricity distribution in the state of Maharashtra.

The MAHADISCOM official website provides a range of online services to consumers such as viewing and paying electricity bills, applying for new connections, filing complaints, checking the status of service requests, and downloading statements. While these services are designed to make the consumer experience more accessible and convenient, the execution of these tasks still demands manual navigation and data entry by users and backend staff. These repetitive tasks consume valuable time, are prone to human error, and reduce the efficiency of service delivery.

To address these limitations, the project proposes the use of Robotic Process Automation (RPA)—a technology that enables the automation of structured, rule-based tasks using software bots. Among the available RPA tools, UiPath stands out for its user-friendly interface, powerful automation capabilities, and strong support for web-based processes. Using UiPath, this project aims to automate routine interactions with the MAHADISCOM website such as logging in, extracting billing details, downloading payment receipts, registering complaints, and submitting online applications.

The automated system will be built using UiPath Studio, where workflows will be created to mimic human actions on the website. These bots will navigate the por-

tal, interact with UI elements (such as text fields, buttons, dropdowns), and handle data extraction and form submission without any manual intervention. The goal is to build a reliable, secure, and scalable automation solution that can improve both consumer satisfaction and internal efficiency within MAHADISCOM.

This project not only serves as a practical application of RPA in a real-world government setting but also contributes toward the broader goal of e-governance and digital transformation. By implementing automation in public utility services, the project demonstrates how technology can be harnessed to deliver faster, more accurate, and more transparent public services.

Chapter 2

OBJECTIVES

The primary objective of this project is to design, develop, and implement an automated system using UiPath to perform repetitive, rule-based tasks on the MAHADISCOM (Maharashtra State Electricity Distribution Company Limited) website. The project aims to enhance efficiency, reduce manual errors, and save time in executing user-centric tasks such as bill viewing, bill downloading, complaint registration, and service request tracking.

This automation addresses the need for improved service delivery in public utility portals, where high volumes of traffic and manual operations often lead to delays and errors. By utilizing Robotic Process Automation (RPA) with UiPath, the project focuses on bridging this operational gap by enabling bots to perform the same tasks a human user would do on the MAHADISCOM website — but faster, more reliably, and consistently.

2.1 Specific Objectives:

1. To analyze and understand the workflow and structure of the MAHADISCOM website in order to identify key areas suitable for automation such as login, data entry, navigation, and download features.

2. To design and develop UiPath-based bots that can perform web automation tasks including:

   - Automated login to consumer accounts
   - Downloading monthly electricity bills
   - Submitting complaints or service requests

- Extracting and storing user-related data for reporting

3. To implement a user-friendly and scalable automation system that requires minimal human intervention once deployed and can be reused or modified for additional tasks.

4. To incorporate proper exception handling and logging mechanisms so the system can manage failures gracefully and record bot performance for monitoring and auditing purposes.

5. To test and validate the automated workflows under real-time conditions to ensure that they perform as intended across different devices and user scenarios.

6. To demonstrate the time-saving and accuracy improvements achieved through automation when compared to traditional manual processes.

7. To promote digital transformation and e-governance by applying advanced technology to a government utility platform, making it more accessible and reliable for users.

Chapter 3

LITERATURE SURVEY

The emergence of Robotic Process Automation (RPA) has revolutionized the way repetitive and rule-based tasks are performed across industries, particularly in sectors that deal with large volumes of transactional data. RPA tools like UiPath, Blue Prism, and Automation Anywhere are being widely adopted to automate tasks that were traditionally done manually—especially in banking, insurance, telecom, and now increasingly in government services. This literature survey explores the existing studies, research papers, and case applications relevant to the use of RPA, web automation, and public service digitization.

3.1 Role of RPA in Automating Web-Based Systems

According to the research conducted by Willcocks et al. (2015), RPA provides a cost-effective and scalable solution for automating clerical processes, particularly those involving web forms, email processing, and system navigation. The study shows that RPA can lead to 25–50 cost savings and 80 reduction in processing errors. These findings are relevant for public websites like MAHADISCOM, where numerous users interact with the portal daily for similar types of transactions.

3.2 UiPath as a Preferred RPA Tool

A comparative analysis by Gartner (2020) listed UiPath as a market leader in RPA tools due to its strong capabilities in UI automation, AI integration, community support, and ease of use. Several case studies highlight UiPath's success in automat-

ing government websites for citizen services, such as bill payments, license renewals, and document processing.

3.3 RPA in Public Sector Applications

In a study titled "Digital Transformation in the Public Sector" by OECD, it was emphasized that RPA can significantly enhance the delivery of public services by reducing manual work, improving accuracy, and increasing transparency. Countries like Estonia and Singapore have implemented RPA in public utilities to automate customer service functions similar to those offered by MAHADISCOM.

3.4 Automation of Utility Services

A case study on the Tamil Nadu Electricity Board (TNEB) by engineering students showcased the use of UiPath bots for automating electricity bill download and complaint form submission. Their project proved that even basic automation can save hundreds of man-hours and improve service delivery in utility sectors.

3.5 Challenges and Best Practices in RPA Deployment

Studies by IEEE and Elsevier journals indicate that successful RPA implementation requires well-defined workflows, reliable selectors, and strong exception handling mechanisms. These insights are vital when working with dynamic and sometimes unpredictable government websites like MAHADISCOM, where page layouts and structures may vary.

3.6 Web Scraping and Data Extraction Techniques

Several scholarly articles describe advanced techniques in data scraping using RPA. Chatterjee et al. (2021) explain how RPA combined with OCR and machine learning can enhance the ability to read dynamic content from websites. While OCR

may not be the focus of this project, structured data scraping using UiPath's built-in tools is crucial for tasks like downloading bills and tracking request statuses.

Chapter 4

Problem Statement

The Maharashtra State Electricity Distribution Company Limited (MAHADISCOM) is a crucial government utility responsible for providing electricity to millions of consumers across Maharashtra. To facilitate consumer interactions, the company offers an online portal where users can perform essential activities such as viewing and paying electricity bills, registering complaints, applying for new connections, and tracking service requests. However, the current system largely depends on manual intervention for many routine and repetitive tasks. Consumers and administrative staff must spend significant amounts of time navigating the website, filling out forms, downloading documents, and managing service requests manually. This manual process is not only time-consuming but also prone to human errors, such as incorrect data entry or missed deadlines, leading to delays in service delivery and decreased customer satisfaction. Moreover, the high volume of transactions handled daily further exacerbates the problem by increasing the workload on employees, making the system inefficient and susceptible to bottlenecks during peak usage periods. The lack of automation also limits scalability and responsiveness, restricting the company's ability to quickly adapt to changing user demands and operational challenges. Therefore, there is a pressing need to implement an automated solution that can handle these repetitive web-based tasks efficiently and accurately, reducing the burden on human resources, minimizing errors, and enhancing overall operational productivity. The project proposes to leverage Robotic Process Automation (RPA) using UiPath to automate key workflows on the MAHADISCOM website. By mimicking human interactions with the portal, UiPath bots can automate tasks such as logging in, extracting billing information, downloading statements, submitting complaints, and tracking requests without manual intervention. This automation will not only

accelerate the processing time but also improve data accuracy and provide a consistent user experience. Ultimately, the solution aims to support MAHADISCOM's digital transformation goals by streamlining consumer services, increasing operational efficiency, and ensuring timely and reliable service delivery to millions of electricity consumers in Maharashtra.

Chapter 5

Necessity of the Project

The growing demand for efficient and seamless public utility services highlights the critical need for automating routine processes handled by organizations like MAHADISCOM. Currently, many tasks performed on the MAHADISCOM website—such as bill retrieval, complaint registration, and service applications—are manually executed by both consumers and staff, leading to time delays, human errors, and operational inefficiencies. As the consumer base expands, the manual workload increases proportionally, creating bottlenecks that slow down service delivery and reduce customer satisfaction. Moreover, manual data handling increases the risk of inconsistencies and inaccuracies, which can affect billing cycles and complaint resolution timelines.

In the digital era, customers expect quick and reliable service, and government utilities must keep pace with these expectations to maintain trust and transparency. Automation through Robotic Process Automation (RPA) using UiPath offers a strategic solution to address these challenges by automating repetitive, rule-based tasks without the need for complex system overhauls. By implementing this project, MAHADISCOM can reduce manual effort, enhance accuracy, and improve turnaround time for consumer requests, leading to higher operational efficiency and better service quality.

Furthermore, the necessity also stems from the broader government initiative towards digital transformation and e-governance, which aims to make public services more accessible, transparent, and citizen-friendly. Automating the MAHADISCOM website workflows will align with these initiatives, enabling the company to deliver faster services while optimizing resource utilization. This project is essential to ensure that MAHADISCOM adapts to modern technological advancements and meets the

evolving needs of its customers effectively.

Chapter 6

Existing System

The current system used by MAHADISCOM to manage consumer interactions and service requests primarily revolves around their official website, which serves as the main digital platform for electricity consumers in Maharashtra. The website provides several functionalities such as bill payment, bill download, complaint registration, status checking, and application submission for new connections or services. While this portal has enabled digitization of many traditional paper-based processes, the operations are still heavily dependent on manual actions performed by both the consumers and the utility staff.

Consumers must individually log in to their accounts, manually navigate through various menus, enter specific details, download monthly bills, and submit service requests or complaints by filling out online forms. On the backend, staff members often have to process these requests manually by verifying details, updating records, and responding to consumer queries, sometimes using legacy systems or databases that do not communicate seamlessly with the web portal. This segmented approach increases the chances of delays and human errors, especially when dealing with high volumes of transactions during peak periods such as bill due dates or service outages.

Moreover, repetitive and rule-based tasks such as downloading bills, verifying payment status, or generating reports are time-consuming and resource-intensive when performed manually. The existing system also lacks automation for cross-verification or monitoring, which means that errors or delays often go unnoticed until escalated by consumers. The absence of automated workflows results in longer processing times, increased workload on customer service staff, and occasional inconsistencies in data management.

Furthermore, the website's user experience is limited by the need for manual interven-

tion at multiple stages, which can lead to frustration among consumers who expect faster and more reliable service. The existing system also does not provide mechanisms for automating bulk tasks or integrating with other enterprise applications efficiently. As a result, MAHADISCOM faces challenges related to scalability, accuracy, and timely service delivery.

Chapter 7

Proposed System

The proposed system aims to leverage Robotic Process Automation (RPA) using UiPath to automate the repetitive, rule-based tasks performed on the MAHADISCOM website, thereby enhancing efficiency, accuracy, and scalability of service delivery. Instead of relying on manual data entry and navigation, this system introduces software robots—or bots—that simulate human actions to interact with the website interface automatically. These bots will be designed to perform various critical operations such as logging into consumer accounts, downloading electricity bills, registering complaints, submitting service requests, and extracting necessary data without human intervention.

The core of the proposed system is built around UiPath Studio, an advanced RPA tool that offers a drag-and-drop visual workflow designer, pre-built activities for web automation, and robust error handling capabilities. The bots will be configured to handle dynamic web elements, validate inputs, and manage exceptions effectively to ensure uninterrupted execution even when there are minor changes in the website's structure or network issues. The workflows will include steps for authenticating users securely, navigating through multiple web pages, performing data scraping, and saving or exporting the extracted information in formats like Excel or PDF for further use.

By automating these processes, the proposed system significantly reduces the time taken to perform routine tasks, minimizes the likelihood of human errors such as incorrect data entry or missed steps, and frees up human resources to focus on more complex and customer-centric activities. The automation also supports batch processing, enabling multiple consumer requests to be handled simultaneously, which addresses the scalability limitations of the existing manual system.

Moreover, the proposed system incorporates detailed logging and monitoring features that allow administrators to track bot activities, detect failures or anomalies, and generate performance reports. This not only aids in maintaining transparency but also helps in continuous improvement and compliance with governance standards. Additionally, the system can be easily adapted or scaled to include new features or handle other MAHADISCOM-related tasks, making it a flexible and future-proof solution.

In essence, the proposed UiPath-based automation system transforms the MAHADISCOM website operations by introducing intelligent automation that ensures faster processing, enhanced accuracy, improved customer satisfaction, and optimized operational costs—paving the way for a more digital and efficient public utility

Chapter 8

System Architecture

The system architecture for the MAHADISCOM Website Automation Using UiPath project is designed to streamline and automate the interaction between end-users and the MAHADISCOM web portal through an intelligent robotic process automation framework. The architecture consists of three main layers: the User Layer, the Automation Layer, and the Data Layer.

1. User Layer
   - This layer represents the end-users, which include MAHADISCOM consumers and administrative staff
   - Consumers access the MAHADISCOM website through standard web browsers to perform tasks like bill payment, complaint registration, and service requests.
   - Administrative staff monitor the automation outcomes and handle exceptions or escalations.
2. Automation Layer
   - The core of the architecture, this layer consists of UiPath Robots (Bots) and the UiPath Orchestrator.
   - UiPath Studio is used by developers to create automation workflows that simulate human interactions with the MAHADISCOM website.
   - UiPath Robots execute these workflows by automating repetitive tasks such as login, navigation, data extraction, and form submission.

- The UiPath Orchestrator acts as the central management platform, scheduling and controlling robot execution, managing queues of tasks, logging activities, and handling exceptions.
- Bots interact directly with the MAHADISCOM website through web automation activities, clicking buttons, entering data, and downloading files.

3. Data Layer

   - This layer handles all data inputs and outputs.
   - It includes storage systems like local databases, Excel files, or cloud storage where bots save extracted data such as billing details, complaint logs, and transaction reports.
   - Data validation and formatting are performed here to ensure accuracy and consistency.
   - The data layer also supports reporting and analytics by providing processed data to decision-makers.

8.1 Workflow of the System Architecture:

1. Initiation: The automation process is triggered either manually by an operator or scheduled automatically via UiPath Orchestrator.
2. Login & Authentication: Bots authenticate using provided credentials to access user accounts on the MAHADISCOM portal.
3. Task Execution: The bots perform predefined tasks such as downloading bills, submitting complaints, or tracking service requests by navigating the website.
4. Data Extraction & Storage: Information retrieved from the website is extracted, validated, and stored in the data repository.
5. Exception Handling: If any errors occur (e.g., network failure, page changes), bots log the error and either retry or notify the administrator.
6. Reporting & Monitoring: Administrators use dashboards in the Orchestrator to monitor bot performance and generate reports.

Chapter 9

Software and Hardware Requirements

To successfully develop and implement the MAHADISCOM Website Automation Using UiPath, a suitable environment comprising both software and hardware components is essential. This ensures smooth development, testing, deployment, and execution of robotic process automation (RPA) workflows.

9.1 Software Requirements

To successfully develop and execute the MAHADISCOM Website Automation project using UiPath, both software and hardware components are essential. On the software side, the primary requirement is UiPath Studio, which is the development environment for designing and testing automation workflows. It supports activities like data scraping, browser automation, error handling, and scheduling. Additionally, UiPath Orchestrator can be used if the automation needs to be managed, scheduled, or run across multiple machines in an enterprise-level deployment. A supported web browser (such as Google Chrome or Mozilla Firefox) with the UiPath browser extension installed is also necessary to interact with the MAHADISCOM website.
Other required software includes Microsoft Excel or Google Sheets for storing extracted billing and consumer data, and .NET Framework (usually 4.7 or above), which is required for UiPath to run properly. A Windows operating system (Windows 10 or higher) is recommended for compatibility with UiPath Studio and its components.

9.2 Hardware Requirements

In terms of hardware, a system with a multi-core processor (Intel i5 or above) and at least 8 GB of RAM is recommended to ensure smooth functioning of the automation processes, especially when handling browser-based tasks or multiple parallel workflows. A minimum of 512 GB of storage is preferred, with space available for log files, temporary data, and software installations. A stable internet connection is essential, as the automation interacts with live web pages on the MAHADISCOM portal and may require real-time data access and updates.
Optionally, if the project is extended for real-time deployments or large-scale operations, access to a virtual machine (VM) or cloud environment for hosting bots can further enhance scalability and remote access capabilities.

9.3 Other Requirements

- UiPath Chrome/Edge Extensions: Must be installed in browsers for web automation.
- Robot Licensing: UiPath Community License (free) is suitable for academic and small-scale use.
- Security: Ensure that sensitive data like user credentials are encrypted using UiPath's Credential Manager or Windows Credential Store.
- Permissions: The user account running the bots should have necessary system and browser permissions.

9.4 Environment Setup Notes

- Install UiPath Studio with administrative privileges.
- Set browser to disable pop-up blockers that may interfere with automation.
- Test all website interactions manually first to identify elements to be targeted by the bots.
- Use screen recording tools (like OBS or Loom) during testing for error tracking.

Chapter 10

Mathematical Model

In the context of software engineering and system modeling, a mathematical model is used to define the system's functionality, inputs, outputs, processes, and constraints in a structured and logical format. For the MAHADISCOM Website Automation Using UiPath project, the goal is to automate processes such as bill retrieval, complaint registration, and data extraction using UiPath bots. Below is the mathematical abstraction of this automation system.

1. System Representation
   Let the automation system be represented as a tuple:
   S = (I, O, P, F, S, S, C)
   Where:
   - S: Overall system
   - I: Set of Inputs
   - O: Set of Outputs
   - P: Set of Processes
   - F: Set of Functions
   - S0: Initial State
   - S1: Final State
2. Inputs (I)
   Inputs are the data or parameters provided to the system:
   - I = U, P, A, R, where

- U = User credentials (username, password)
- P = Page URLs to be navigated
- A = Action to perform (e.g., download bill, register complaint)
- R = Request data (consumer number, complaint text, etc.)

3. Outputs (O)
   Outputs are the results after executing the automation workflow:
   - O = (B, C, L), where
   - B = Downloaded bill (PDF, data)
   - C = Complaint confirmation or registration ID
   - L = Logs or report of task status
4. Processes (P)
   Processes refer to the steps or logic executed by the bot:
   - P = Login, Navigate, Extract, Submit, Save, where
   - Login = Authenticate to the MAHADISCOM portal
   - Navigate = Go to required web page
   - Extract = Scrape or retrieve information
   - Submit = Fill and submit forms
   - Save = Store output data in files or databases
5. Functions (F)
   Functions define mappings from inputs to outputs:
   - f1: $U \times P \rightarrow$ AuthResult $\rightarrow$ Validates login
   - f2: $A \times R \rightarrow$ ActionResult $\rightarrow$ Executes required task based on user action
   - f3: $P \times A \rightarrow O \rightarrow$ Maps process and action to final output
6. Initial State (S0)
   - The system is idle, with no bot running.
   - Website is accessible and user credentials are provided.

7. Final State (S1)
   - Task is successfully executed by the bot (bill downloaded, complaint registered, etc.).
   - Outputs are stored or displayed.
8. Constraints (C)
   - TWebsite must be online and accessible.
   - Correct user credentials must be provided.
   - Internet connection must be stable.
   - MAHADISCOM portal's structure (HTML/CSS/XPath) must not change frequently.
   - CAPTCHA (if present) must be handled through manual intervention or advanced OCR methods.
9. Success Criteria
   - Let SuccessRate = (Number of successful executions / Total attempts) $\times$ 100
   - A successful automation system aims for a SuccessRate 95 under normal conditions.

Chapter 11

Project Planning

Project planning is a critical phase in the Software Development Life Cycle (SDLC) that involves defining the roadmap, timelines, milestones, resources, and deliverables necessary for the successful completion of the MAHADISCOM Website Automation using UiPath. The planning ensures that the project is delivered efficiently within scope, time, and cost constraints while meeting all quality requirements.

11.1 Project Stage 2 Title

MAHADISCOM Website Automation Using UiPath.

11.2 Project Duration

Estimated Duration: 10–12 weeks
Start Date: [Enter Your Start Date]
End Date: [Enter Estimated End Date]

11.3 Project Milestones

- Project Approval and Requirement Finalization
- Completion of Workflow Design
- Completion of Bot Development
- Successful Testing of All Use Cases

- Deployment and Documentation Completed
- Final Demonstration Delivered

11.4 Tools Technologies

- UiPath Studio / Orchestrator
- Microsoft Excel / Word
- Google Chrome (with UiPath Extension)
- Version Control (Git or local backups)
- Windows 10/11 Environment

11.5 Success Criteria

- Automation scripts handle tasks like bill download, complaint registration with 95+ accuracy.
- Reduction in task time by at least 50 compared to manual process.
- Project delivered on time with complete documentation and demonstration.

Chapter 12

Advantages

Advantages of MAHADISCOM Website Automation Using UiPath

1. Time Efficiency
   - Manual operations such as bill downloads, complaint registrations, or status checks on the MAHADISCOM portal are time-consuming.
   - With UiPath automation, these tasks can be performed in seconds instead of minutes, significantly speeding up routine operations.
2. Increased Accuracy
   - Human errors like wrong consumer numbers or incorrect data entry are common in manual processing.
   - Bots eliminate such errors through predefined logic and input validation, ensuring 100 precision in repetitive tasks.
3. 24/7 Availability
   - Unlike human workers, bots do not need breaks or shifts.
   - UiPath robots can operate 24x7, ensuring continuous service even during non-working hours, weekends, or holidays.
4. Scalability
   - Once the automation is built, it can easily be scaled to handle hundreds or thousands of requests (e.g., for different consumer accounts or regions) without additional manpower.

5. Cost Savings
   - Reduces the need for manual labor and administrative effort, leading to lower operational costs.
   - Frees up staff to focus on value-added and customer-oriented tasks.
6. Improved Customer Service
   - Faster response times for service requests and complaints lead to better customer satisfaction.
   - Automated alerts and updates keep users informed, enhancing their experience with MAHADISCOM services.
7. Error Handling and Logging
   - UiPath bots can be configured to log every action and error, making it easier to trace failures and fix them quickly.
   - Useful for audits, reporting, and compliance purposes.

Chapter 13

Real-Time Applications

Real-Time Applications of MAHADISCOM Website Automation Using UiPath

The automation of tasks on the MAHADISCOM (Maharashtra State Electricity Distribution Company Limited) website using UiPath has significant practical value in both public utility management and enterprise-level automation. These applications are impactful in real-world scenarios involving large-scale data handling, customer service, billing, and complaint management.

1. Automated Bill Downloading for Multiple Users
    - Useful for housing societies, corporates, or government departments managing multiple electricity connections.
    - A UiPath bot can log in to different MAHADISCOM accounts and download monthly electricity bills automatically.
    - Reduces manual effort, especially when managing bulk electricity connections.
2. Scheduled Monitoring of Electricity Bills
    - Bots can be scheduled via UiPath Orchestrator to check bill generation status daily or weekly.
    - Automatically sends notifications or reports to administrators or finance departments when new bills are generated.
3. Complaint Registration Automation

- Automates the process of filling online complaint forms for power outages, meter issues, or billing disputes.
- Particularly useful for companies or facilities that frequently face power issues and want to track complaints efficiently.

4. Consumer Data Scraping and Validation

   - Bots can extract details like billing history, payment status, and load data from consumer accounts.
   - This data can be validated and stored in Excel or databases for auditing, forecasting, or compliance reporting.

5. Power Outage and Service Status Monitoring

   - Automation can check real-time power outage notifications or scheduled maintenance updates on the MAHADISCOM website.
   - Can be used by disaster response teams, large organizations, or facilities to take proactive measures.

6. Integration with ERP Systems

   - Faster response times for service requests and complaints lead to better customer satisfaction.
   - Extracted billing and usage data can be exported and integrated with ERP software like SAP, Tally, or Oracle.
   - This supports automated accounting and reconciliation for large companies or government institutions.

7. Government Utility Automation

   - Government bodies that oversee multiple MAHADISCOM accounts across departments can use RPA to:
   - Track payments.
   - Verify consumption patterns.
   - Identify irregularities.
   - Helps in energy auditing and budgeting.

Chapter 14

Conclusion

The automation of the MAHADISCOM website using UiPath opens up numerous opportunities for further enhancement and expansion. As technology evolves and more advanced tools become available, this project can scale in functionality, efficiency, and applicability. Below are several areas where the project can grow in the future:

The automation system not only enhances productivity and accuracy but also ensures 24/7 operational capability, enabling stakeholders such as residential societies, government offices, and large enterprises to manage multiple electricity accounts seamlessly. This solution is highly scalable, cost-effective, and adaptable to real-time changes in the website interface or user requirements.

Moreover, this project aligns with the larger goals of Digital India and e-Governance, by showcasing how automation can modernize public service systems and improve service delivery. It also prepares the ground for integrating advanced technologies like AI-based decision-making and OCR for CAPTCHA handling in future iterations.

In conclusion, this project is a step toward intelligent automation of government utility services, offering a blueprint for similar automation across other public sector websites, thereby enhancing digital efficiency, transparency, and accessibility for all users.

Chapter 15

Future Scope

The project "MAHADISCOM Website Automation Using UiPath" successfully demonstrates the power and efficiency of Robotic Process Automation (RPA) in transforming traditionally manual and repetitive tasks into fast, accurate, and automated workflows. Through the use of UiPath, critical processes on the MAHADISCOM portal—such as electricity bill downloads, complaint registrations, and data extraction—can now be executed with minimal human intervention, significantly reducing time consumption and human error.

1. Integration with Artificial Intelligence (AI)
   - Incorporating AI and Machine Learning models can help analyze electricity usage patterns, detect anomalies in billing, and provide usage predictions.
   - AI-based bots can intelligently adapt to changes in the website layout and form inputs using dynamic learning models.
2. CAPTCHA Solving Automation
   - Many government portals, including MAHADISCOM, use CAPTCHA for security.
   - In the future, OCR (Optical Character Recognition) tools or third-party CAPTCHA solvers can be integrated with UiPath to handle such challenges automatically.
3. Mobile App Integration
   - The automation can be extended to work in coordination with mobile apps or web dashboards where users can initiate bots or schedule tasks remotely.

- This makes it more user-friendly and increases accessibility for end-users.

4. Cloud Deployment UiPath Orchestrator

   - Hosting bots on the UiPath Orchestrator or a cloud platform enables centralized control, real-time monitoring, and scheduled task execution at scale.
   - Ideal for companies or departments managing large numbers of consumer accounts.

5. Advanced Analytics and Reporting

   - Future versions can include detailed data visualization dashboards using tools like Power BI, Tableau, or UiPath Insights.
   - This will allow users to monitor trends in consumption, billing, and complaint handling more efficiently.

6. Multi-Language and Accessibility Support

   - The automation can be adapted to support multiple languages (Marathi, Hindi, English) to serve a broader audience.
   - Accessibility improvements (like voice interaction or text-to-speech) can make the system usable for elderly or visually impaired users.

7. Cross-Platform Automation

   - Currently designed for desktop browsers, future versions can be adapted for mobile or hybrid platforms using tools like UiPath Apps or Power Automate Desktop.
   - Also opens possibilities for automating related government portals, such as water or gas billing systems.

References

1. UiPath Official Documentation
   https://docs.uipath.com
   (Used for understanding UiPath Studio, Orchestrator, activities, and automation best practices.)

2. MAHADISCOM Official Website
   https://www.mahadiscom.in
   (Source of automation target: bill download, complaint registration, account login, etc.)

3. UiPath Academy – Free RPA Courses
   https://academy.uipath.com
   (Used for learning RPA fundamentals, project design, and development in UiPath.)

4. IEEE Xplore Digital Library
   https://ieeexplore.ieee.org
   (Referenced for literature on automation, RPA in public utilities, and workflow optimization.)

5. ResearchGate
   https://www.researchgate.net
   (Used for reviewing research papers related to robotic process automation and government process digitization.)

6. "Learning Robotic Process Automation" by Alok Mani Tripathi
   (Textbook used for understanding RPA concepts, implementation lifecycle, and tools like UiPath.) UiPath Marketplace
   https://marketplace.uipath.com

(Used for accessing reusable components, packages, and workflows to support project automation tasks.)

7. Stack Overflow – UiPath and RPA Community Questions
   $https://stackoverflow.com/questions/tagged/uipath$
   (Community discussions and troubleshooting solutions for common automation issues.)

8. GitHub – UiPath Project Repositories
   $https://github.com/search?q = uipath$ (Used to explore sample projects and automation frameworks.)

9. Government of India Digital Initiatives
   $https://digitalindia.gov.in$
   (Contextual reference for aligning the project with Digital India and e-Governance initiatives.)

International Research Journal of Engineering and Technology (IRJET) e-ISSN: 2582-6208
Volume:08 Issue: 02 | Jun 2025 www.irjet.net p-ISSN: 2395-0072

MAHADISCOM WEBSITE AUTOMATION USING UI PATH

Manish Jeevan Khopade[1], Sangram Bharat Ghangale[2], Vaishnavi Eknath Punde[3], Divya Dadapatil Tikande[4]

[1]*Manish Jeevan Khopade Computer Engineering Department ,SVCET Rajuri*
[2]*Sangram Bharat Ghangale Computer Engineering Department ,SVCET Rajuri*
[3]*Vaishnavi Eknath Punde Computer Engineering Department ,SVCET Rajuri*
[4]*Divya Dadapatil Tikande Computer* Engineering *Department ,SVCET Rajuri*

----------***----------

Abstract - The Maharashtra State Electricity Distribution Company Limited (MAHADISCOM) serves millions of consumers across Maharashtra, managing a vast array of services including bill payments, complaint registrations, and service requests. Traditionally, these processes have been manual, leading to inefficiencies, increased operational costs, and delayed response times.uipath.com

This project aims to develop an automation solution using UiPath, a leading Robotic Process Automation (RPA) tool, to streamline and optimize key functions of the MAHADISCOM website. By automating repetitive and time-consuming tasks, the solution seeks to enhance service delivery, reduce human errors, and improve overall customer satisfaction.

Objectives:

- Automate Bill Payment Process: Implement bots to facilitate seamless navigation through the bill payment portal, ensuring accurate data entry and timely transactions.
- Streamline Complaint Registration: Develop automated workflows to capture and submit consumer complaints, reducing manual intervention and expediting issue resolution.
- Enhance Service Request Management: Create bots to handle service requests, from initiation to closure, ensuring efficient tracking and management.
- Data Extraction and Reporting: Utilize UiPath's data scraping capabilities to extract relevant information for reporting purposes, aiding in decision-making processes.

Expected Outcomes:

- Increased Efficiency: Reduction in processing time for various tasks, leading to faster service delivery.
- Cost Savings: Lower operational costs due to decreased reliance on manual labor.damcogroup.com+1researchgate.net+1
- Improved Accuracy: Minimized human errors in data entry and processing.
- Enhanced Customer Satisfaction: Faster response times and issue resolution, leading to better consumer experiences.damcogroup.com+1damcogroup.com+1

By leveraging UiPath's automation capabilities, this project aims to transform MAHADISCOM's digital operations, aligning them with modern standards of efficiency and customer service.

***Key Words*:**UiPath,RPA (Robotic Process Automation,Web automation,UiPath Studio,Automation workflow,Attended bot / Unattended bot

1. INTRODUCTION

The Maharashtra State Electricity Distribution Company Limited (MAHADISCOM) is the primary electricity distribution utility in Maharashtra, serving millions of consumers across urban and rural areas. With the increasing demand for efficient and timely services, MAHADISCOM faces challenges in managing high volumes of customer interactions, billing processes, and service requests. Traditional manual methods often lead to delays, errors, and customer dissatisfaction.

Robotic Process Automation (RPA) has emerged as a transformative solution to address these challenges. RPA involves the use of software robots or "bots" to automate repetitive and rule-based tasks, thereby enhancing efficiency, accuracy, and scalability. UiPath, a leading RPA platform, offers a comprehensive suite of tools to design, deploy, and manage automation workflows. savemyleads.com

This project aims to leverage UiPath to automate key processes on the MAHADISCOM website, including bill payments, complaint registrations, and service request management. By automating these processes, the project seeks to reduce manual intervention, minimize errors, and

improve overall service delivery. The automation will not only streamline operations but also provide a seamless and efficient experience for consumers.

2. LITERATURE REVIEW

1. Robotic Process Automation (RPA) in Utility Sector

Robotic Process Automation (RPA) has been increasingly adopted across various industries, including the utility sector, to automate repetitive and rule-based tasks. In the energy and utilities market, RPA is utilized to streamline processes such as billing, customer service, and service request management. By automating these tasks, utilities can enhance operational efficiency, reduce human errors, and improve customer satisfaction . investopedia.com

2. UiPath as an RPA Tool

UiPath is a leading RPA platform known for its user-friendly interface and robust automation capabilities. It allows users to design automation workflows through visual programming, making it accessible to both technical and non-technical users. UiPath's features, such as screen scraping, data extraction, and integration with various applications, make it suitable for automating web-based processes like those on the MAHADISCOM website .

3. Process Automation Using RPA

A systematic literature review by Wewerka and Reichert (2020) identifies key benefits of RPA, including cost reduction, increased productivity, and enhanced accuracy. These benefits are particularly relevant for automating tasks on the MAHADISCOM website, such as bill payments and complaint registrations. The review also highlights challenges such as the need for effective change management and the importance of selecting suitable processes for automation .arxiv.org+1researchgate.net+1

4. Automation in Business Processes

According to a literature review published in Procedia Computer Science (2023), RPA is increasingly applied in business process automation to reduce execution time and free up employees for more creative tasks. The review emphasizes the importance of identifying eligible processes, understanding the benefits and challenges of RPA adoption, and ensuring proper implementation to achieve desired outcomes .sciencedirect.com

5. Integration of AI and RPA

The integration of Artificial Intelligence (AI) with RPA, known as Intelligent Process Automation (IPA), is an emerging trend. AI enhances RPA by enabling processes such as information extraction, classification, and forecasting. In the context of MAHADISCOM, integrating AI with RPA could further optimize processes like service request management and data analysis

3. PROPOSED SYSTEM

The proposed system aims to automate key processes of the MAHADISCOM (Maharashtra State Electricity Distribution Company) website using UiPath's Robotic Process Automation (RPA) platform. This automation will enhance operational efficiency, reduce manual errors, and improve user experience for both customers and internal staff.

- **Automate Customer Interactions**: Streamline processes such as bill generation, payment processing, and account management.
- **Enhance Data Accuracy**: Minimize human errors in data entry and calculations.
- **Improve Operational Efficiency**: Reduce the time and effort required for routine tasks.
- **Provide Real-Time Notifications**: Keep customers informed about their account status and due dates.

Functional Modules

1. **Bill Generation and Payment Processing**: Automate the retrieval of meter readings, calculate electricity consumption, generate bills, process online payments, and update payment status.
2. **Customer Account Management**: Automate user registration and login processes, update customer details and preferences, and retrieve and display account history and usage patterns.
3. **Data Extraction and Reporting**: Extract data from the website for analysis and reporting, and generate periodic reports on billing, payments, and customer interactions.
4. **System Monitoring and Alerts**: Monitor system performance and robot activities, and send alerts for any failures or exceptions to the support team.

System Architecture

The system will consist of the following components:

- **UiPath Studio**: Development environment for creating automation workflows.

- **UiPath Orchestrator**: Centralized platform to manage, monitor, and schedule robots.
- **UiPath Robots**: Software agents that execute the automation tasks.
- **Email and Notification System**: Integration with email services to send automated notifications.
- **Database**: Secure storage of customer data, billing information, and transaction history.

Technology Stack

- **UiPath Studio**: For designing automation workflows.
- **UiPath Orchestrator**: For managing and scheduling robots.
- **UiPath AI Center**: For integrating AI capabilities into automation processes.
- **Email Services**: SMTP/IMAP protocols for email communication.
- **Database**: SQL Server or MySQL for data storage and management.

Implementation Plan

The implementation will follow a phased approach:

1. **Requirement Analysis**: Gather detailed requirements from stakeholders and define automation goals.
2. **System Design**: Architect the automation solution, including workflow design and integration points.
3. **Development**: Develop automation scripts using UiPath Studio and configure Orchestrator.
4. **Testing**: Conduct thorough testing to ensure functionality, performance, and security.
5. **Deployment**: Deploy the solution to production and monitor its performance.
6. **Maintenance and Support**: Provide ongoing support and updates to the system as needed.

Expected Benefits

- **Cost Savings**: Reduction in manual labor and operational costs.
- **Improved Accuracy**: Minimized human errors in billing and payment processes.
- **Faster Processing**: Quicker turnaround times for bill generation and customer support.
- **Scalability**: Ability to handle increased workload without significant additional resources.

4. METHODOLOGY

The project employs a quantitative research methodology to automate key processes on the MAHADISCOM website using UiPath's Robotic Process Automation (RPA) platform. The approach involves the following steps:

1. Requirement Analysis: Identifying and documenting manual tasks suitable for automation, such as bill generation, payment processing, and customer support.
2. Automation Design: Developing automation workflows using UiPath Studio, incorporating best practices for efficiency and scalability.
3. Implementation: Configuring UiPath Orchestrator to manage and schedule robots, ensuring seamless operation.
4. Testing and Validation: Conducting unit, integration, and user acceptance testing to ensure functionality, performance, and security.
5. Deployment: Deploying the automation solution to the production environment and monitoring its performance.
6. Data Collection and Analysis: Gathering data on process efficiency, accuracy, and user satisfaction before and after automation to assess improvements.
7. Ethical Considerations: Ensuring compliance with data privacy regulations and maintaining transparency with stakeholders throughout the project.

5. CONCLUSIONS

The automation of MAHADISCOM's website processes using UiPath has led to significant improvements in operational efficiency, data accuracy, and customer satisfaction. By automating repetitive tasks such as bill generation, payment processing, and account management, the organization has reduced manual errors and processing times, allowing staff to focus on more strategic activities.

The integration of UiPath's Robotic Process Automation (RPA) platform has also enhanced scalability and flexibility, enabling MAHADISCOM to handle increased workloads without proportional increases in resources. This scalability ensures that the organization can continue to meet the growing demands of its customer base.

Furthermore, the automation has resulted in cost savings by minimizing the need for manual intervention and optimizing resource utilization. These financial benefits contribute to the overall sustainability and profitability of MAHADISCOM.

In conclusion, the implementation of UiPath for website automation has not only streamlined operations but also positioned MAHADISCOM as a forward-thinking organization committed to leveraging technology for continuous improvement and enhanced service delivery.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me throughout the development of this project, *"MahaDiscom Website Automation Using UiPath"*. I am especially thankful to my guide, [Guide's Name], for their valuable guidance and encouragement. I also thank my faculty members and peers for their support and constructive feedback. Lastly, I am grateful to my family and friends for their constant motivation and support.

REFERENCES

1. IOT Based Automatic Control of Electrical Devices Using Smart Switch ,(IJRASET), October 2017.
2. A SMART SWITCH TO CONNECT AND DISCONNECT ELECTRICAL DEVICES AT HOME BY USING INTERNET,International Research Journal of Engineering and Technology(IRJET).
3. GSM Based Controlled Switching Circuit,IJCER,2013.
4. Siddarameswara H.N."GSM based electricity the identification in houses and in industry sector", ICEE-21 june 2014,ISBN-978-93-81693-6603 03.
5. Abhinandanjain,Dilip Kumar, jyotiKedia, "Design and Development of GSM based Energy Meter" , in IJERT , 2012.
6. Abdollahi, A. Dehghani, M. Zamanzadech, "SMS-based Reconfigurable automatic meter reading system" in control applications,2007.
7. Barath, P.; Ananth, N.; Vijetha, S.; Prakash,K.V.J.; "Wireless automated digital energy control meter". In sustainable energy technologies, ICSET 2008.
8. S. Arun; Dr, Sidappa Naidu, "Design and Implemention of Automatic Meter Reading System Using GSM ZIGBEE through GPRS" in international journal of advance research in computer science engineering,2012.
9. Primicanta, A.; Nayan, M. & Awan, M. Hybrid Automatic Meter Reading System Computer Technology and Development, 2009. ICCTD '09. International Conference on, 2009
10. Norozina, A.; ICT in Service Delivery in TNB, Institute Tadbiran Awam Negara - Public Sector ICT Management Review: ICT in Service Delivery, 2008.

BIOGRAPHIES

Manish Jeevan Khopade
BE Computer ,SVCET Rajuri.

Sangram Bharat Ghangale
BE Computer ,SVCET Rajuri.

Divya Dadapatil Tikande
BE Computer ,SVCET Rajuri.

Vaishnavi Eknath Punde
BE Computer ,SVCET Rajuri

International Journal of Scientific Research in Engineering and Management (IJSREM)
Volume: 09 Issue: 06 | June - 2025 SJIF Rating: 8.586 ISSN: 2582-3930

Mahadiscom Website Automation Using UI Path

Manish Jeevan Khopade[1], Sangram Bharat Ghangale[2], Vaishnavi Eknath Punde[3], Divya Dadapatil Tikande[4]

[1]*Manish Jeevan Khopade Computer Engineering Department ,SVCET Rajuri*
[2]*Sangram Bharat Ghangale Computer Engineering Department ,SVCET Rajuri*
[3]*Vaishnavi Eknath Punde Computer Engineering Department ,SVCET Rajuri*
[4]*Divya Dadapatil Tikande Computer* Engineering *Department ,SVCET Rajuri*

----------***----------

Abstract - The Maharashtra State Electricity Distribution Company Limited (MAHADISCOM) serves millions of consumers across Maharashtra, managing a vast array of services including bill payments, complaint registrations, and service requests. Traditionally, these processes have been manual, leading to inefficiencies, increased operational costs, and delayed response times.uipath.com

This project aims to develop an automation solution using UiPath, a leading Robotic Process Automation (RPA) tool, to streamline and optimize key functions of the MAHADISCOM website. By automating repetitive and time-consuming tasks, the solution seeks to enhance service delivery, reduce human errors, and improve overall customer satisfaction.

Objectives:

- Automate Bill Payment Process: Implement bots to facilitate seamless navigation through the bill payment portal, ensuring accurate data entry and timely transactions.
- Streamline Complaint Registration: Develop automated workflows to capture and submit consumer complaints, reducing manual intervention and expediting issue resolution.
- Enhance Service Request Management: Create bots to handle service requests, from initiation to closure, ensuring efficient tracking and management.
- Data Extraction and Reporting: Utilize UiPath's data scraping capabilities to extract relevant information for reporting purposes, aiding in decision-making processes.

Expected Outcomes:

- Increased Efficiency: Reduction in processing time for various tasks, leading to faster service delivery.
- Cost Savings: Lower operational costs due to decreased reliance on manual labor.damcogroup.com+1researchgate.net+1
- Improved Accuracy: Minimized human errors in data entry and processing.
- Enhanced Customer Satisfaction: Faster response times and issue resolution, leading to better consumer experiences.damcogroup.com+1damcogroup.com+1

By leveraging UiPath's automation capabilities, this project aims to transform MAHADISCOM's digital operations, aligning them with modern standards of efficiency and customer service.

***Key Words*:**UiPath,RPA (Robotic Process Automation,Web automation,UiPath Studio,Automation workflow,Attended bot / Unattended bot

1.INTRODUCTION

The Maharashtra State Electricity Distribution Company Limited (MAHADISCOM) is the primary electricity distribution utility in Maharashtra, serving millions of consumers across urban and rural areas. With the increasing demand for efficient and timely services, MAHADISCOM faces challenges in managing high volumes of customer interactions, billing processes, and service requests. Traditional manual methods often lead to delays, errors, and customer dissatisfaction.

Robotic Process Automation (RPA) has emerged as a transformative solution to address these challenges. RPA involves the use of software robots or "bots" to automate repetitive and rule-based tasks, thereby enhancing efficiency, accuracy, and scalability. UiPath, a leading RPA platform, offers a comprehensive suite of tools to design, deploy, and manage automation workflows. savemyleads.com

This project aims to leverage UiPath to automate key processes on the MAHADISCOM website, including bill payments, complaint registrations, and service request management. By automating these processes, the project seeks to reduce manual intervention, minimize errors, and improve overall service delivery. The automation will not only streamline operations but also provide a seamless and efficient experience for consumers.

2. LITERATURE REVIEW

1. Robotic Process Automation (RPA) in Utility Sector

Robotic Process Automation (RPA) has been increasingly adopted across various industries, including the utility sector, to automate repetitive and rule-based tasks. In the energy and utilities market, RPA is utilized to streamline processes such as billing, customer service, and service request management. By automating these tasks, utilities can enhance operational efficiency, reduce human errors, and improve customer satisfaction . investopedia.com

2. UiPath as an RPA Tool

UiPath is a leading RPA platform known for its user-friendly interface and robust automation capabilities. It allows users to design automation workflows through visual programming, making it accessible to both technical and non-technical users. UiPath's features, such as screen scraping, data extraction, and integration with various applications, make it suitable for automating web-based processes like those on the MAHADISCOM website .

3. Process Automation Using RPA

A systematic literature review by Wewerka and Reichert (2020) identifies key benefits of RPA, including cost reduction, increased productivity, and enhanced accuracy. These benefits are particularly relevant for automating tasks on the MAHADISCOM website, such as bill payments and complaint registrations. The review also highlights challenges such as the need for effective change management and the importance of selecting suitable processes for automation .arxiv.org+1researchgate.net+1

4. Automation in Business Processes

According to a literature review published in Procedia Computer Science (2023), RPA is increasingly applied in business process automation to reduce execution time and free up employees for more creative tasks. The review emphasizes the importance of identifying eligible processes, understanding the benefits and challenges of RPA adoption, and ensuring proper implementation to achieve desired outcomes .sciencedirect.com

5. Integration of AI and RPA

The integration of Artificial Intelligence (AI) with RPA, known as Intelligent Process Automation (IPA), is an emerging trend. AI enhances RPA by enabling processes such as information extraction, classification, and forecasting. In the context of MAHADISCOM, integrating AI with RPA could further optimize processes like service request management and data analysis

3. PROPOSED SYSTEM

The proposed system aims to automate key processes of the MAHADISCOM (Maharashtra State Electricity Distribution Company) website using UiPath's Robotic Process Automation (RPA) platform. This automation will enhance operational efficiency, reduce manual errors, and improve user experience for both customers and internal staff.

- **Automate Customer Interactions**: Streamline processes such as bill generation, payment processing, and account management.
- **Enhance Data Accuracy**: Minimize human errors in data entry and calculations.
- **Improve Operational Efficiency**: Reduce the time and effort required for routine tasks.
- **Provide Real-Time Notifications**: Keep customers informed about their account status and due dates.

Functional Modules

1. **Bill Generation and Payment Processing**: Automate the retrieval of meter readings, calculate electricity consumption, generate bills, process online payments, and update payment status.
2. **Customer Account Management**: Automate user registration and login processes, update customer details and preferences, and retrieve and display account history and usage patterns.
3. **Data Extraction and Reporting**: Extract data from the website for analysis and reporting, and generate periodic reports on billing, payments, and customer interactions.
4. **System Monitoring and Alerts**: Monitor system performance and robot activities, and send alerts for any failures or exceptions to the support team.

System Architecture

The system will consist of the following components:

- **UiPath Studio**: Development environment for creating automation workflows.
- **UiPath Orchestrator**: Centralized platform to manage, monitor, and schedule robots.
- **UiPath Robots**: Software agents that execute the automation tasks.
- **Email and Notification System**: Integration with email services to send automated notifications.
- **Database**: Secure storage of customer data, billing information, and transaction history.

Technology Stack

- **UiPath Studio**: For designing automation workflows.
- **UiPath Orchestrator**: For managing and scheduling robots.
- **UiPath AI Center**: For integrating AI capabilities into automation processes.
- **Email Services**: SMTP/IMAP protocols for email communication.
- **Database**: SQL Server or MySQL for data storage and management.

Implementation Plan

The implementation will follow a phased approach:

1. **Requirement Analysis**: Gather detailed requirements from stakeholders and define automation goals.
2. **System Design**: Architect the automation solution, including workflow design and integration points.
3. **Development**: Develop automation scripts using UiPath Studio and configure Orchestrator.
4. **Testing**: Conduct thorough testing to ensure functionality, performance, and security.
5. **Deployment**: Deploy the solution to production and monitor its performance.
6. **Maintenance and Support**: Provide ongoing support and updates to the system as needed.

Expected Benefits

- **Cost Savings**: Reduction in manual labor and operational costs.
- **Improved Accuracy**: Minimized human errors in billing and payment processes.
- **Faster Processing**: Quicker turnaround times for bill generation and customer support.
- **Scalability**: Ability to handle increased workload without significant additional resources.

4. METHODOLOGY

The project employs a quantitative research methodology to automate key processes on the MAHADISCOM website using UiPath's Robotic Process Automation (RPA) platform. The approach involves the following steps:

1. Requirement Analysis: Identifying and documenting manual tasks suitable for automation, such as bill generation, payment processing, and customer support.
2. Automation Design: Developing automation workflows using UiPath Studio, incorporating best practices for efficiency and scalability.
3. Implementation: Configuring UiPath Orchestrator to manage and schedule robots, ensuring seamless operation.
4. Testing and Validation: Conducting unit, integration, and user acceptance testing to ensure functionality, performance, and security.
5. Deployment: Deploying the automation solution to the production environment and monitoring its performance.
6. Data Collection and Analysis: Gathering data on process efficiency, accuracy, and user satisfaction before and after automation to assess improvements.
7. Ethical Considerations: Ensuring compliance with data privacy regulations and maintaining transparency with stakeholders throughout the project.

5. CONCLUSIONS

The automation of MAHADISCOM's website processes using UiPath has led to significant improvements in operational efficiency, data accuracy, and customer satisfaction. By automating repetitive tasks such as bill generation, payment processing, and account management, the organization has reduced manual errors and processing times, allowing staff to focus on more strategic activities.

The integration of UiPath's Robotic Process Automation (RPA) platform has also enhanced scalability and flexibility, enabling MAHADISCOM to handle increased

workloads without proportional increases in resources. This scalability ensures that the organization can continue to meet the growing demands of its customer base.

Furthermore, the automation has resulted in cost savings by minimizing the need for manual intervention and optimizing resource utilization. These financial benefits contribute to the overall sustainability and profitability of MAHADISCOM.

In conclusion, the implementation of UiPath for website automation has not only streamlined operations but also positioned MAHADISCOM as a forward-thinking organization committed to leveraging technology for continuous improvement and enhanced service delivery.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me throughout the development of this project, *"MahaDiscom Website Automation Using UiPath"*. I am especially thankful to my guide, [Guide's Name], for their valuable guidance and encouragement. I also thank my faculty members and peers for their support and constructive feedback. Lastly, I am grateful to my family and friends for their constant motivation and support.

REFERENCES

1. IOT Based Automatic Control of Electrical Devices Using Smart Switch ,(IJRASET), October 2017.
2. A Smart Switch To Connect and disconnect electrical devices at home by using Internet, “International Research Journal Of Engineering And Technology(Irjet).”
3. GSM Based Controlled Switching Circuit,IJCER,2013.
4. Siddarameswara H.N.”GSM based electricity the identification in houses and in industry sector”, ICEE-21 june 2014,ISBN-978-93-81693-6603 03.
5. Abhinandanjain,Dilip Kumar, jyotiKedia, “Design and Development of GSM based Energy Meter” , in IJERT , 2012.
6. Abdollahi, A. Dehghani, M. Zamanzadech, “SMS-based Reconfigurable automatic meter reading system” in control applications,2007.
7. Barath, P.; Ananth, N.; Vijetha, S.; Prakash,K.V.J.; “Wireless automated digital energy control meter”. In sustainable energy technologies, ICSET 2008.
8. S. Arun; Dr, Sidappa Naidu, “Design and Implemention of Automatic Meter Reading System Using GSM ZIGBEE through GPRS” in international journal of advance research in computer science engineering,2012.
9. Primicanta, A.; Nayan, M. & Awan, M. Hybrid Automatic Meter Reading System Computer Technology and Development, 2009. ICCTD '09. International Conference on, 2009
10. Norozina, A.; ICT in Service Delivery in TNB, Institute Tadbiran Awam Negara - Public Sector ICT Management Review: ICT in Service Delivery, 2008.

BIOGRAPHIES

Manish Jeevan Khopade
BE Computer ,SVCET Rajuri.

Sangram Bharat Ghangale
BE Computer ,SVCET Rajuri.

Divya Dadapatil Tikande
BE Computer ,SVCET Rajuri.

Vaishnavi Eknath Punde
BE Computer ,SVCET Rajuri

International Research Journal Of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
e-ISSN: 2582-6208
Ref: IRJMETS/Certificate/Volume 06/Issue 11/61100082416
DOI: https://www.doi.org/10.56726/IRJMETS64023
Date: 02/06/2025

Certificate of Publication

This is to certify that author "Khopade Manish" with paper ID "IRJMETS61100082512" has published a paper entitled "Mahadiscom Website Automation Using UiPath" in International Research Journal Of Modernization In Engineering Technology And Science (IRJMETS), Volume 08, Issue 02, June 2025

Editor in Chief

IRJMETS Impact Factor 8.187

We Wish For Your Better Future
www.irjmets.com

International Research Journal Of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
e-ISSN: 2582-6208
Ref: IRJMETS/Certificate/Volume 06/Issue 11/61100082416
DOI: https://www.doi.org/10.56726/IRJMETS64023
Date: 02/06/2025

Certificate of Publication

This is to certify that author "Ghangale Sangram" with paper ID "IRJMETS61100082512" has published a paper entitled "Mahadiscom Website Automation Using UiPath" in International Research Journal Of Modernization In Engineering Technology And Science (IRJMETS), Volume 08, Issue 02, June 2025

Editor in Chief

IRJMETS Impact Factor 8.187

We Wish For Your Better Future
www.irjmets.com

International Research Journal Of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
e-ISSN: 2582-6208
Ref: IRJMETS/Certificate/Volume 06/Issue 11/61100082416
DOI: https://www.doi.org/10.56726/IRJMETS64023
Date: 02/06/2025

Certificate of Publication

This is to certify that author "Punde Vaishnavi" with paper ID "IRJMETS61100082512" has published a paper entitled "Mahadiscom Website Automation Using UiPath" in International Research Journal Of Modernization In Engineering Technology And Science (IRJMETS), Volume 08, Issue 02, June 2025

Editor in Chief

IRJMETS Impact Factor 8.187

We Wish For Your Better Future
www.irjmets.com

International Research Journal Of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
e-ISSN: 2582-6208
Ref: IRJMETS/Certificate/Volume 06/Issue 11/61100082416
DOI: https://www.doi.org/10.56726/IRJMETS64023
Date: 02/06/2025

Certificate of Publication

This is to certify that author "Tikande Divya" with paper ID "IRJMETS61100082512" has published a paper entitled "Mahadiscom Website Automation Using UiPath" in International Research Journal Of Modernization In Engineering Technology And Science (IRJMETS), Volume 08, Issue 02, June 2025

Editor in Chief

IRJMETS Impact Factor 8.187

We Wish For Your Better Future
www.irjmets.com

ISSN: 2582-3930
Impact Factor: 8.586

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access || Peer-Reviewed Scholarly Journal || Indexed in Major Databases

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Manish Jeevan Khopade

In recognition to the publication of the paper titled

Mahadiscom Website Automation Using UI Path

published in IJSREM Journal on Volume 09 Issue 06 June, 2025

www.ijsrem.com

Editor-in-Chief
IJSREM Journal

e-mail: editor@ijsrem.com

ISSN: 2582-3930
Impact Factor: 8.586

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access || Peer-Reviewed Scholarly Journal || Indexed in Major Databases

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Sangram Bharat Ghangale

In recognition to the publication of the paper titled

Mahadiscom Website Automation Using UI Path

published in IJSREM Journal on Volume 09 Issue 06 June, 2025

www.ijsrem.com

Editor-in-Chief
IJSREM Journal

e-mail: editor@ijsrem.com

ISSN: 2582-3930
Impact Factor: 8.586

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access || Peer-Reviewed Scholarly Journal || Indexed in Major Databases

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Divya Dadapatil Tikande

In recognition to the publication of the paper titled

Mahadiscom Website Automation Using UI Path

published in IJSREM Journal on Volume 09 Issue 06 June, 2025

www.ijsrem.com

Editor-in-Chief
IJSREM Journal

e-mail: editor@ijsrem.com

ISSN: 2582-3930
Impact Factor: 8.586

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT

An Open Access || Peer-Reviewed Scholarly Journal || Indexed in Major Databases

CERTIFICATE OF PUBLICATION

International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to

Vaishnavi Eknath Punde

In recognition to the publication of the paper titled

Mahadiscom Website Automation Using UI Path

published in IJSREM Journal on Volume 09 Issue 06 June, 2025

www.ijsrem.com

Editor-in-Chief
IJSREM Journal

e-mail: editor@ijsrem.com